



BIO-CODES

Enhancing AI-Readiness of Bioimaging Data

ISCC Introduction

2024-11-29, *Titusz Pan*

About me | Titusz Pan

- Founder and CEO of Craft AG, Freiburg
- Open-Source Software Developer
- Expert at ISO/TC 46/SC 9/WG 18
- Chairman at non-profit ISCC Foundation



Who is the ISCC Foundation

The ISCC Foundation is a purpose-driven non-profit, dedicated to developing, standardizing and promoting open-source technology for **universal content identification**.



Titusz Pan
Chairman



Kira Lemke
Treasurer



Todd Carpenter
Advisory Board



Roanie Levy
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Giacomo D'Angelo
Advisory Board



Frank Shulleri
Advisory Board



Sebastian Posth
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Lambert Heller
Advisory Board



Philippe Rixhon
Advisory Board



ISCC Foundation Activities



- Content Identification Research
- Open source development
- IT infrastructure for ISCC services
- Promoting ISCC adoption
- Governance of the ISCC



TC 46 - Information and Documentation

SC 09 - Identification and Description

ISCC

ISO 24138:2024

Conceived
2016-06-29

WG 18 Started
2019-10-29

Published
2024-05-15

ISO 2108:2017	ISBN	International Standard Book Number
ISO 3297:2022	ISSN	International Standard Serial Number
ISO 3901:2019	ISRC	International Standard Recording Code
ISO 15706-1:2023	ISAN	International Standard Audiovisual Number
ISO 15707:2022	ISWC	International Standard Musical Work Code
ISO 27729:2012	ISNI	International Standard Name Identifier
ISO 26324:2022	DOI	Digital Object Identifier System
ISO 24138:2024	ISCC	International Standard Content Code



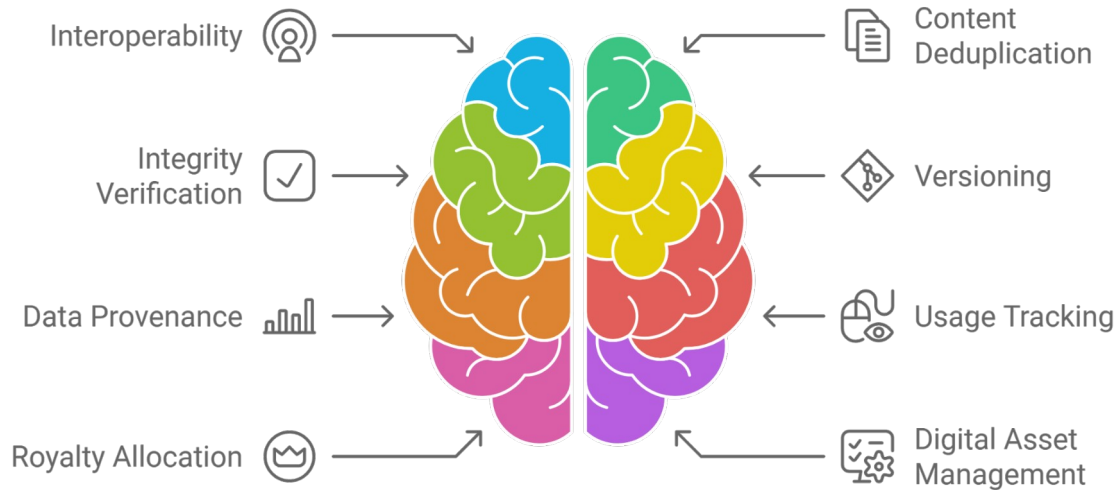
Why is ISO 24138:2024 a significant publication?



- Open-source and interoperable content identification & fingerprinting system
- One universal code for digital text, image, audio and video
- Cross-sector standard (publishing, science, arts, etc.)
- A neutral, transparent and global system

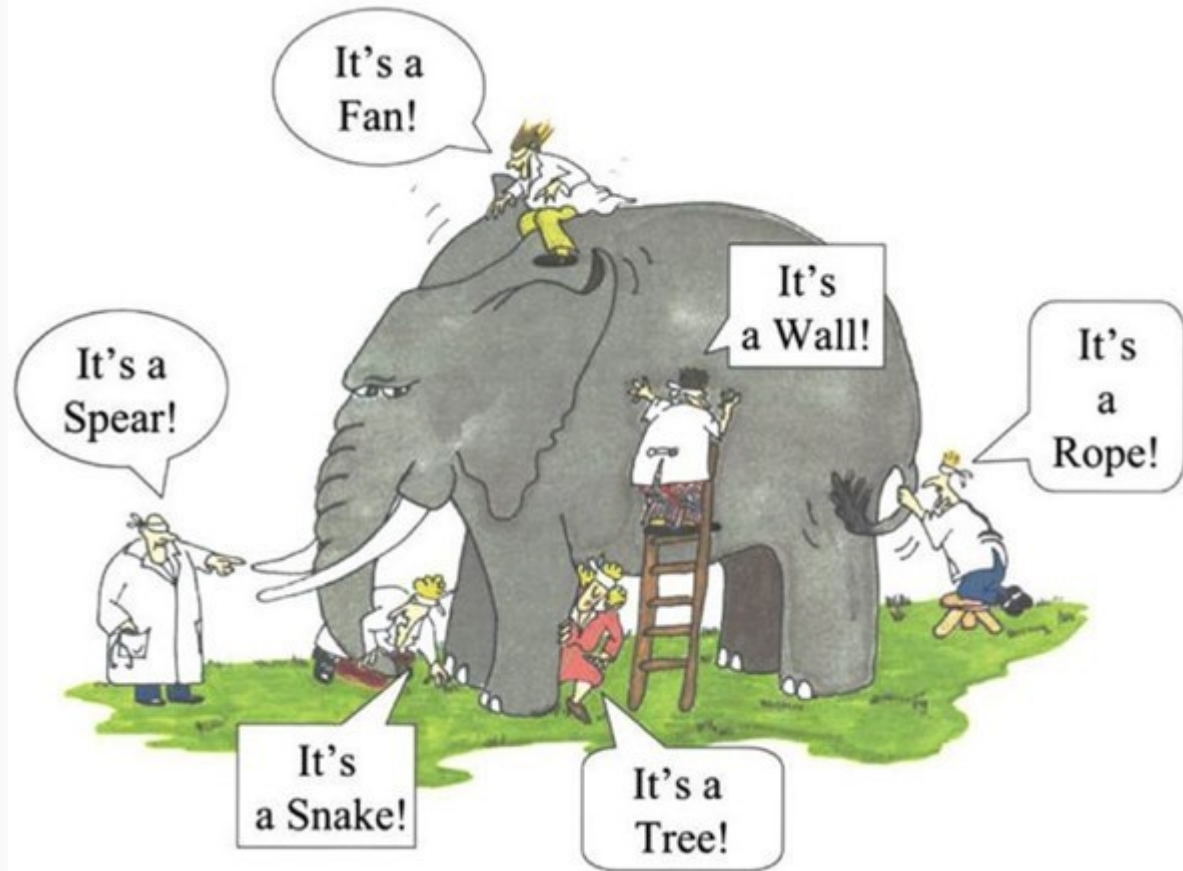


Key Benefits of Implementing ISCC



What is **Not** ISCC

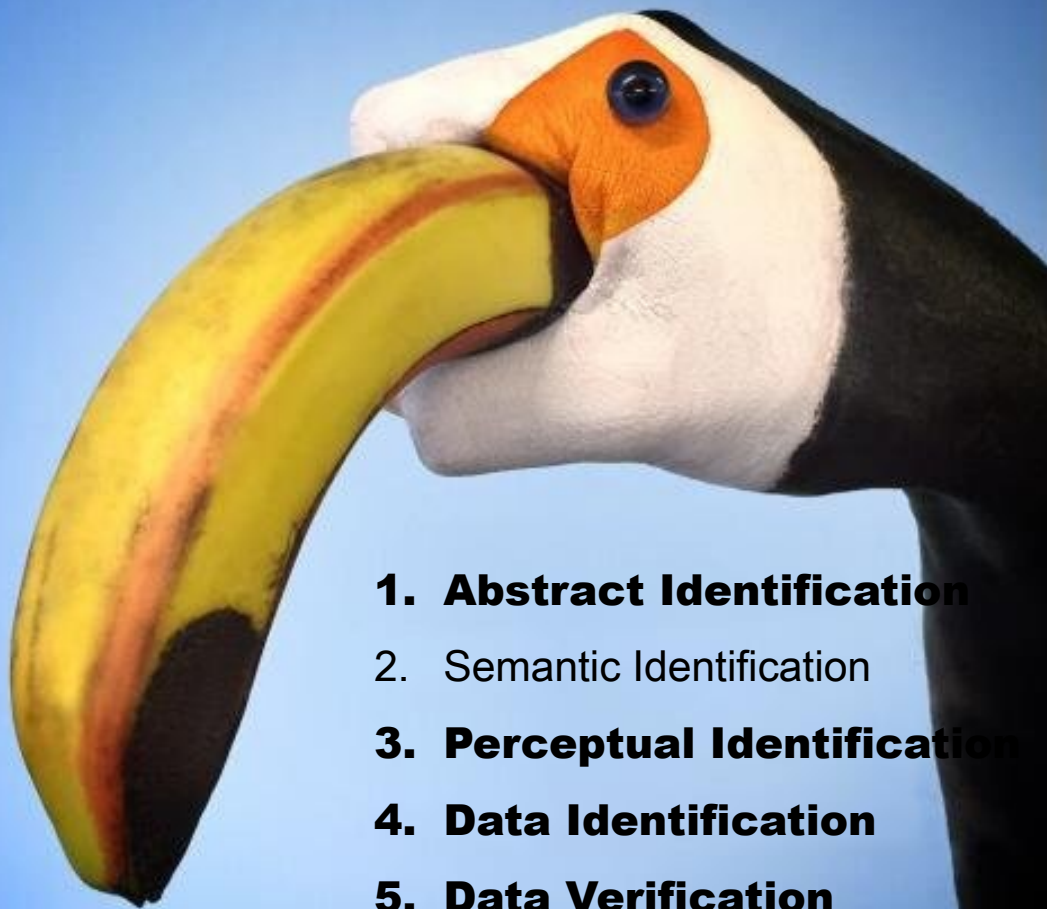
- A persistent identifier
- A content recognition system
- A cryptographic hash
- A content registry



Layers of "Content" Identification

In our model for digital content identification we distinguish 6 layers that exist naturally on a scale from abstract to concrete.

Existing content identifiers usually operate on one or two of these layers.



[Image courtesy of Imgur](#)

- 1. Abstract Identification**
2. Semantic Identification
- 3. Perceptual Identification**
- 4. Data Identification**
- 5. Data Verification**
6. Individual Copy



The DNA of your digital content

Estimate similarity using ISCC-CODEs

ISCC: KED572P4AOF5K6QXQA4T6OJD5UGX7UBPFW2TVQNTHBCKFRFCANCZARQ4K6NSFZQSH
4GQ

Meta-Code

AAA572P4AOF5K6Q
X

Semantic-Code

CEAYAOJ7HER62DL
7

Content-Code

EEA5ALZNWU5MD
MZY

Data-Code

GAAUJIWEUIBULEC
G

Instance-Code

IAARYV43ELTBEPY
N

Abstract & Persistent

Concrete & Volatile

Metadata
Similarity

Semantic
Similarity

Syntactic
Similarity

Data
Similarity

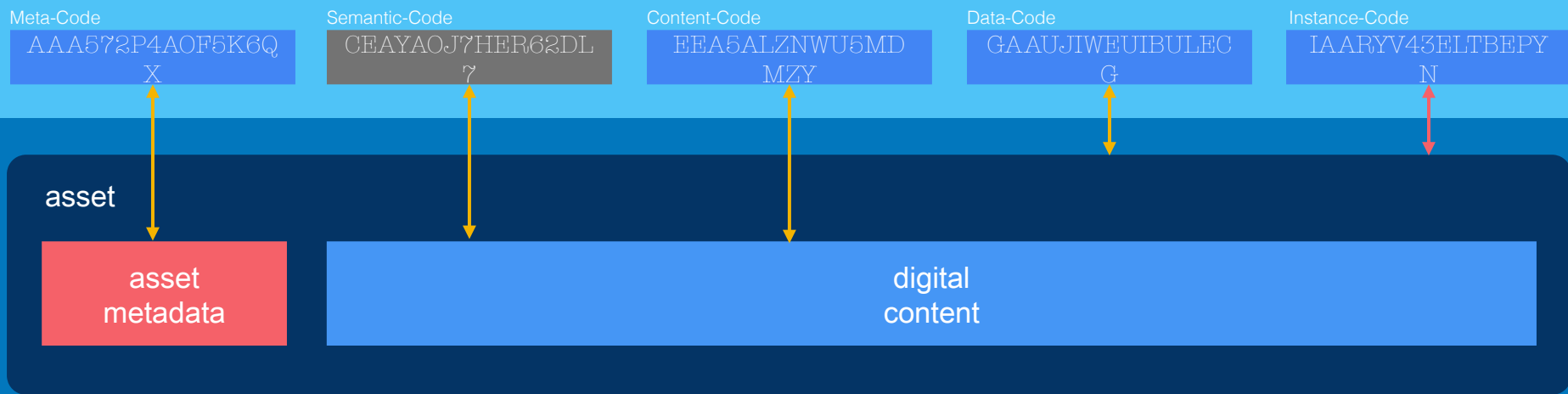
Data
Integrity

Components are self-describing and can be used standalone or in combination and at different length

ISCC & Content Binding

a binding of bindings

ISCC:KED572P4AOF5K6QXQA4T6OJD5UGX7UBPFW2TVQNTHBCKFRFCANCZARQ4K6NSFZQSH
4GQ



Yellow arrows are soft bindings, red arrows are hard bindings.

Additionally ISO 24138:2024 defines a **Meta-Hash** algorithm which is a secure hard binding to be part of ISCC metadata.

CCDFPFo87Mhd
T

CTWAGYJ9HZGj
1

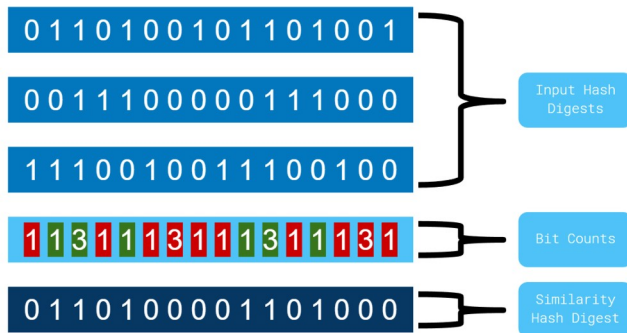
CDhydSjQXDxV
k

CRd5bk4SrEpzt

Meta-Code

A similarity preserving
hash over metadata.

ISCC - Similarity Hash Diagram



Layer 1 - Abstract Identification

The **Meta-Code** is seeded from Metadata

name: Title for content or work or series (max 128 bytes)

description: Description of the identified content (max 4096 bytes)

meta: Subject, industry, or use-case specific metadata, encoded as RFC-2397 Data-URL (max 16384 bytes).

- Identifies at any desired level of abstraction (series, work ...)
- Top level of grouping a content collection or hierarchy
- Independent of digital manifestations
- Supports *progressive disambiguation*
- Should be embedded into digital assets for reproducibility
- Requires minimal metadata

Seed Metadata is metadata that is used to establish a Meta-Code and stays frozen (immutable) throughout its existence. Floating Metadata is any mutable metadata that is managed in context with an **ISCC**.

CCDFPFo87Mhd
TCTWAGYJ9HZGj
1CDhydSjQXDxV
k

CRd5bk4SrEpzt

Layer 3 - Perceptual Identification

Content-Code (Image)

Similarity hash over normalized generic data. Self-Describing and media-type specific.

If we want to identify “Content” we cannot compare on encoded “Data”:

- Two “identical” images
- Yet the data is completely different
- Due to different file formats
- Content-ID encodes information structure - not raw data



=



JPG Data

```
49 74 27 73
20 6e 6f 74
20 61 62 6f
75 74 20 62
61 6e 6b 69
6e 67 20 74
68 65 20 75
6e 62 61 6e
6b 65 64 2e
```

≠

PNG Data

```
54 68 65 20
43 75 72 72
65 6e 63 79
20 75 73 65
64 20 6f 6e
20 43 6f 62
6c 6f 20 69
73 20 43 68
61 72 6d 2e
```

JPG SHA1

```
7b 24 1f 77
f0 f2 96 df
73 b5 e0 38
97 6a 5e 3b
d0 12 bd 23
```

≠

PNG SHA1

```
7e bd c5 c5
c0 30 d5 4c
30 c0 31 df
4c 9e ff d5
b2 ad e8 2d
```

JPG Content-ID

CYHa5UMqq1iQS

=

PNG Content-ID

CYHa5UMqq1iQS



CCDFPFc87Mhd
T

CTWAGYJ9HZGj
1

CDhydSjQXDxV
k

CRd5bk4SrEpzt

Data-Code

Similarity over raw encoded data.

- Identifies encoded content
- Clusters file versions
- Spectrum of tolerance
- Shift resistant chunking (CDC)
- **Similarity hash (tree) over variable sized chunk hashes**

Layer 4 - Encoded Manifestation

Fixed Size Data Chunking

AAAA | BBBB | CCCC | DDDD |

EAAA | BBBB | CCCC | DDDD | D

Content Defined Chunking - Shift Resistant - Variable Size Chunks

AAAA | BBBB | CCCC | DDDD |

EAAAAA | BBBB | CCCC | DDDD |

CCDFPFo87Mhd
T

CTWAGYJ9HZGj
1

CDhydSjQXDXV
k

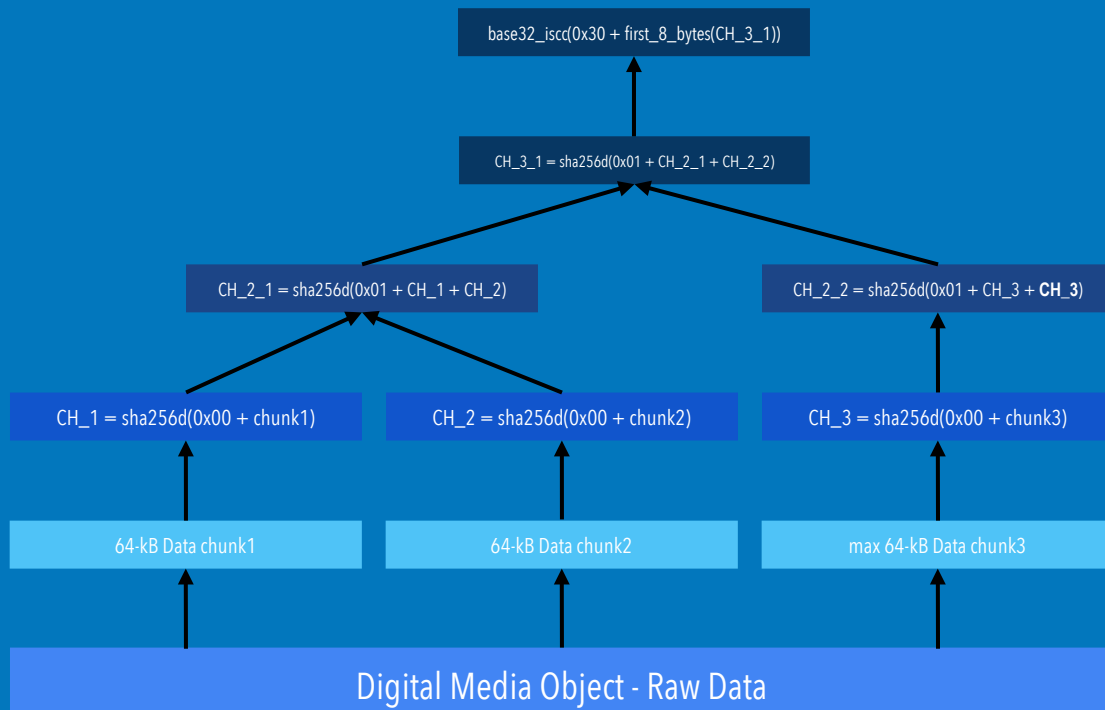
CRd5bk4SrEpzt

Instance-Code

Cryptographic hash. The root of a hash tree over raw data.

- Precise data identification
- Proof of data containment
- Separate Tophash (256 bit)
- Data integrity (via tophash)

Layer 5 - Exact Representation



Example one DOI multiple matching ISCC



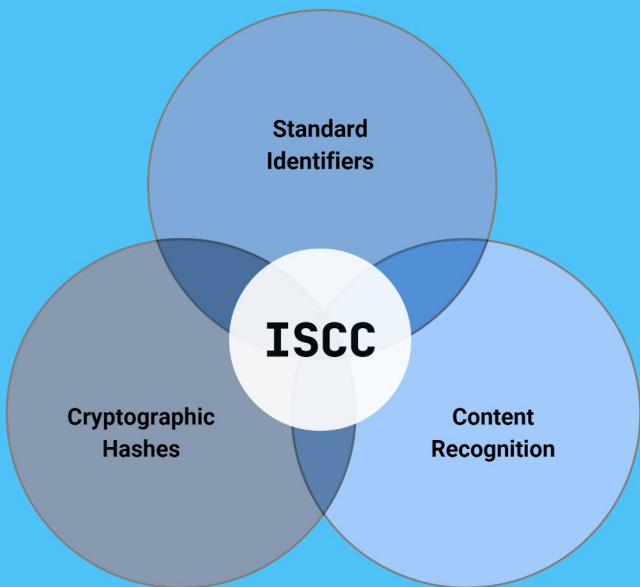
Paper: Neural Computation of Surface Border Ownership and Relative Surface Depth from Ambiguous Contrast Inputs

Host	DOI	ISCC
hal.archives-ouvertes.fr	10.3389/fpsyg.2016.01102	CCDyud5ZWakDR-CTTq25WFQTWaU-CDbUZg6v3qzzM-CRxfuPk2nP3Q
arxiv.org	10.3389/fpsyg.2016.01102	CCDyud5ZWakDR-CTTRs5cQY1D11-CDPqUxrqN7YRx-CRcUmq2SmgN18
hal.archives-ouvertes.fr	10.3389/fpsyg.2016.01102	CCDyud5ZWakDR-CTfNotD3KMMd1-CD481J7LDBQPH-CR8rZ9QzTzJRL
frontiersin.org	10.3389/fpsyg.2016.01102	CCDyud5ZWakDR-CTfNotD3KMMd1-CD481J7LDBQPH-CR8rZ9QzTzJRL

Estimated Similarity of Content-Code Text: 84.38 %
 Estimated Similarity of Data-Code: 53.12 %



At the Intersection of Digital Content Identification



The ISCC combines properties from multiple content identification paradigms:

- **Standard Identifier:**
Unique media asset identification
- **Content Recognition:**
Clusters “*similar*” content
- **Cryptographic Hashes:**
Verifies media asset integrity



Content Binding

Connecting Content with Metadata



01

Standard Identifiers

- Registration based (bound by database entry)
- Abstract “Binding” via descriptive metadata
- Assignment requires third-party

02

Watermark (Soft Binding)

- Hides an identifier within the data (proprietary)
- Can be removed/changed if algorithm is known
- Supports audiovisual content only (no text)

03

Fingerprint (Soft Binding)

- Calculated from the digital asset
- Works for all modalities and cannot be removed
- Not statistically unique

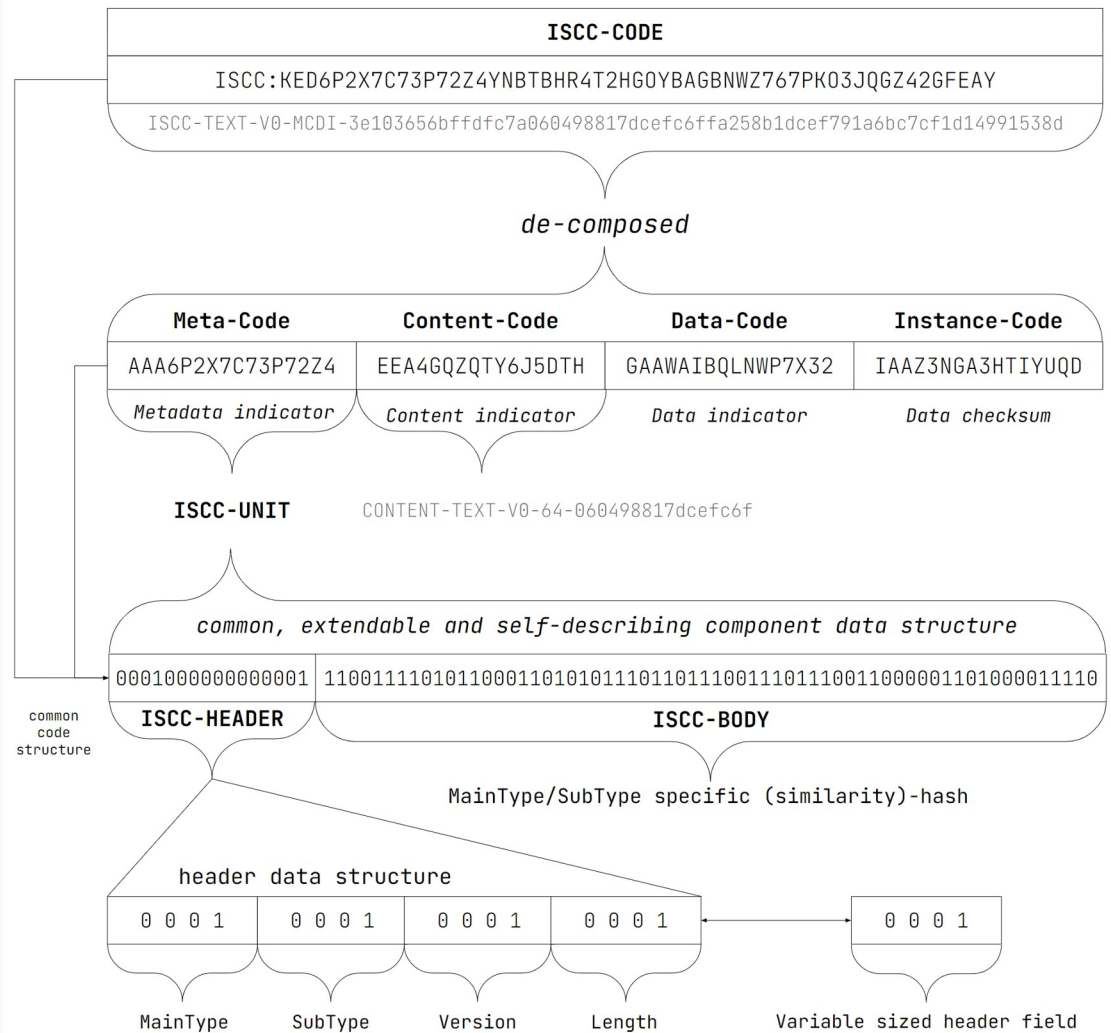
04

Cryptohash (Hard Binding)

- Calculated from the digital asset
- Works for all data
- Strong uniqueness guarantee

ISCC - Codec

The ISCC is based on a compact, machine-readable, self-describing, modular and extensible codec.

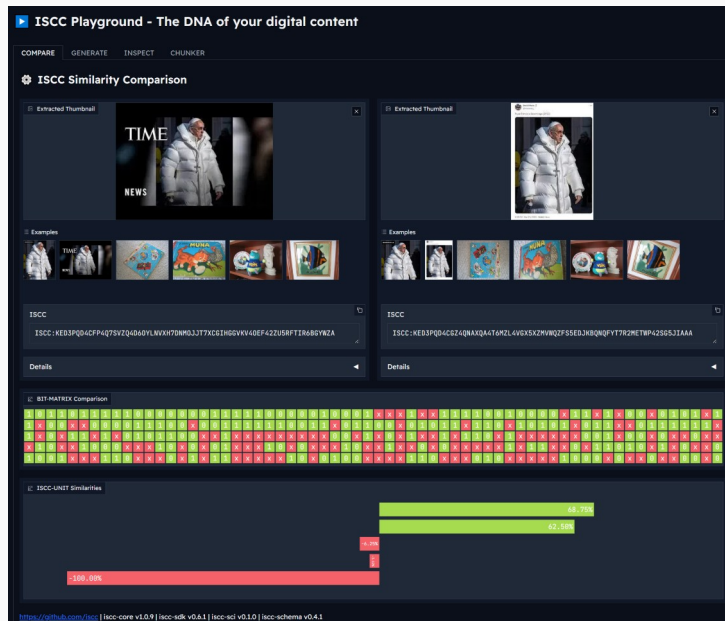


Demo Time :)

Playground



- An interactive playground
- Explore and visualize **ISCC**
- Helps developers and users
- Showcase future updates
- Published on Huggingface



<https://huggingface.co/spaces/iscc/iscc-playground>



Give your DOI superpowers

Bind digital content to a DOI



Timestamps

demonstrate when
your content was
first created

Signatures

cryptographically
prove your work's
authenticity

Fingerprints

help others find your
book metadata and
licensing infos

ISCCs connect your paper to a DOI in a granular, verifiable and location independent way. Individual paragraphs and images can be resolved to a DOI and associated metadata.

Digital Reality

There is too much/granular content to manually assign and track identifiers.

Good News



already has an ISCC.
It just needs to be
computed.



Thank you
for your attention



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